

Class 10th Aarambh 2026-27

DPP #6

#LP 1: Prove that $3 + 2\sqrt{5}$ is irrational.

#LP 2: Two lines are given to be parallel. The equation of one of the lines is $3x - 2y = 5$. The equation of the second line can be

- (a) $9x + 8y = 7$
- (b) $-12x - 8y = 7$
- (c) $-12x + 8y = 7$
- (d) $12x + 8y = 7$

#LP 3: Find HCF and LCM of 404 and 96 and verify that $\text{HCF} \times \text{LCM} =$ Product of the two given numbers.

#LP 4: If α, β are the zeroes of the quadratic polynomial $p(x) = x^2 - (k + 6)x + 2(2k - 1)$, then the value of k , if $\alpha + \beta = \frac{1}{2} \alpha\beta$, is

- (a) -7
- (b) 7
- (c) -3
- (d) 3

#LP 5: Find the smallest value of p for which the quadratic equation $x^2 - 2(p+1)x + p^2 = 0$ has real roots. Hence, find the roots of the equation so obtained.

#LP 6: Check whether the following pair of equations is consistent or not. If consistent, solve graphically.

$$x + 3y = 6$$

$$3y - 2x = -12$$

#LP 7: Find the zeroes of the polynomial $p(x) = 3x^2 - 4x - 4$. Hence, write a polynomial whose each of the zeroes is 2 more than zeroes of $p(x)$.

#LP 8: Find HCF and LCM of 510 and 92 and verify that $\text{HCF} \times \text{LCM} =$ Product of the two given numbers.

Answer Sheet

#LP 1 : $3 + 2\sqrt{5}$ is irrational.

#LP 2 : (c)

#LP 3 : Part I: HCF=4, LCM=9696.

Part II: 38784

#LP 4 : (b)

#LP 5 : $p=-1/2$, $x=1/2, 1/2$

#LP 6 : The pair of eq. is consistent. The sol. obtained graphically is $x=6$, $y=0$.

#LP 7 : Zeros= $-2/3$ and 2.

$$P(x)=3x^2 -16x+16.$$

#LP8: Part I: HCF=2, LCM=23460

Part II: 46920